# COVARIATE SHIFT COMPOUNDING ERROR FLYWHEEL & RECOVERY DYNAMICS

Quadratic Divergence $O(T^2\varepsilon)$ in Naive Behavioral Cloning vs Linear Bounded Tube $O(T\varepsilon)$ in DAgger Corrective Aggregation

## × NAIVE BEHAVIORAL CLONING (OPEN-LOOP DRIFT FLYWHEEL) — Error: $O(T^2\varepsilon)$

**1. Single-Step Error ε**
Friction / noise perturbation

**2. State Exits Support**
s_{t+1} ∉ supp(d^π*)

**3. Arbitrary Action â**
Zero constraint from ERM loss

**4. Amplified Deviation**
Further away on next step

**Mathematical Compounding Mechanism:**

E[Error_BC] ≤ ∑_{t=1}^T t·ε = [T(T+1)/2]·ε = $O(T^2\varepsilon)$

Example (T=500 steps, ε=0.01): Cumulative error multiplier = 2,500×

## ✓ CORRECTIVE AGGREGATION / DAgger (STABILIZING FLYWHEEL) — Error: $O(T\varepsilon)$

**1. Policy Visits s ~ d^π̂**
Learner visits perturbed state

**2. Expert Labels π*(s)**
Recovery action demonstrated

**3. Retrain on Aggregate D**
supp(D) ⊇ recovery envelope

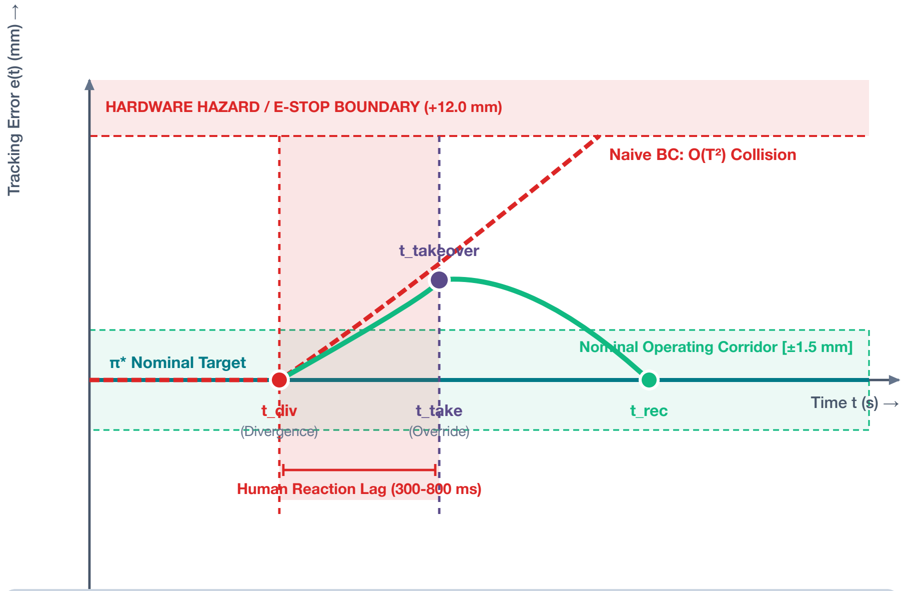
**4. Closed-Loop Recovery**
Corrective torque returns to tube

**Mathematical Linear Bounded Corridor:**

E[Error_CA] ≤ T·ε = $O(T\varepsilon)$ [Ross & Bagnell 2011]

Example (T=500 steps, ε=0.01): Cumulative error multiplier = 5.0× (500× reduction!)

## TRAJECTORY DIVERGENCE & INTERVENTION CUT BOUNDARY




### THE 3 DATA CURATION LAWS FOR PHYSICAL INTERVENTIONS:

**1. Truncate Autonomous Rollout at t_div (NOT t_takeover):**
Cut rollout where policy drifted. Prevents spurious correlation between drift and goal.

**2. Discard Corrupted Drift Interval [t_div, t_takeover]:**
Omit 30-80 execution frames recorded during human perception/reaction delay.

**3. Ingest Recovery Demonstration [t_takeover, t_rec] into DAgger Buffer:**
Treat rescue as expert feedback on policy-induced distribution d^π̂, collapsing error to $O(T\varepsilon)$.